



Outline of the lecture

The history of Informatics & HCI

Our roots and evolution

[NAME_3]

[NAME_4]

[NAME_5]

[NAME_6]

Action Research & [NAME_7]

[NAME_7] (place) studies & ethnography | [NAME_8]

Praxeology & Appropriation | [NAME_9]

Structuration & Infrastructuring

A holistic look at Human-System-Interaction

HCI in [NAME_10]

[NAME_10] | Prototyping | Evaluation

“know the user”

Human computer interaction is a difficult endeavor with glorious rewards.

Designing interactive systems to be

effective

efficient

easy to use

enjoyable to use

*so that **people** and **society** may realize the benefits of computation-based devices*

Human computer interaction is a difficult endeavor with glorious rewards.

Analytic
Design/Implementation

Designing interactive systems to be

effective

efficient

easy to use

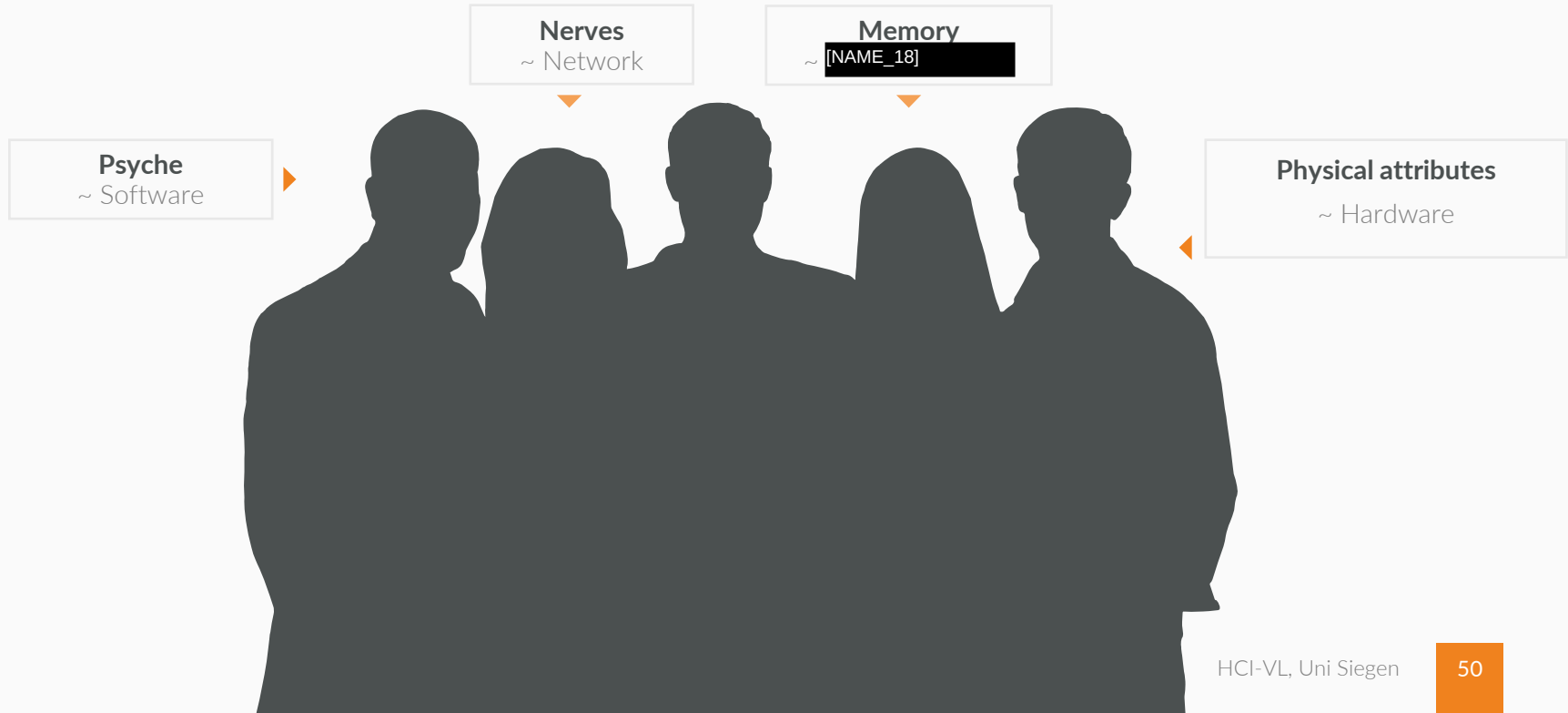
enjoyable to use

*so that **people** and **society** may realize the benefits of computation-based devices*

How to go beyond platitudes like “know the user”?

Humans as [NAME_4] systems

(Questionable) Analogy: Humans as a computer-type system. Exemplary “components” are labelled:



Memory: Encoding, Storage, and

[NAME_14]

[NAME_14]

To Design

[NAME_15]

[NAME_16]

Sensory

[NAME_17]

Perception of the world/system/enviornment

Information entering through the sensory cortices of the brain and relaying through the thalamus. It lasts only milliseconds and is mostly outside conscious awareness.

Working

[NAME_18]

[NAME_17] of attention / Short-term memory

The structures and processes used for temporarily storing and manipulating information

Long-term

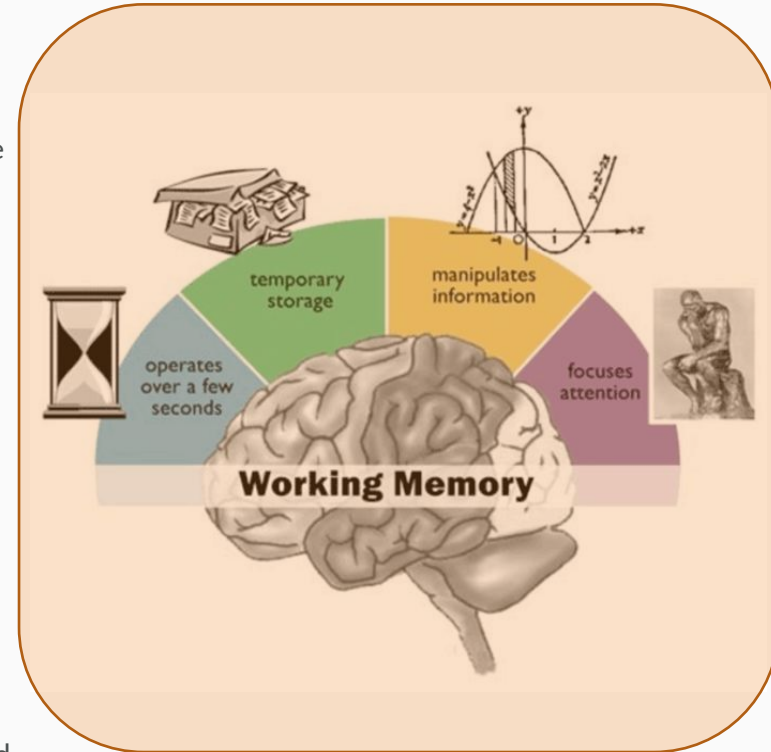
[NAME_21]

[NAME_18] and [NAME_19]

[NAME_19]: Declarative, Episodic, Semantic, [NAME_20]

[NAME_20]: How our perceptions become memories?

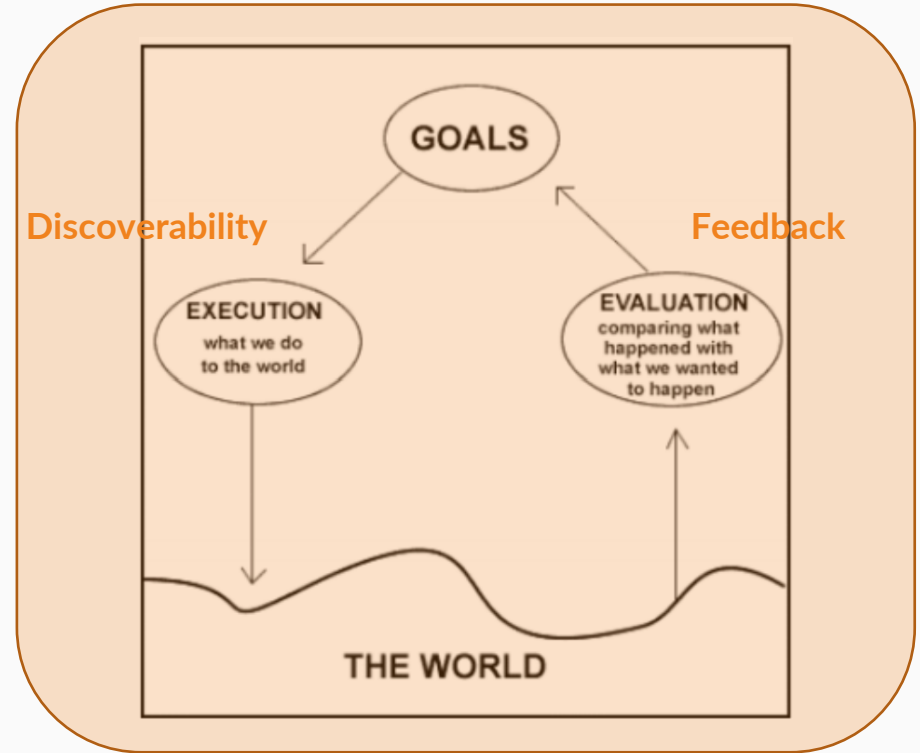
Unless information is encoded, it cannot be remembered



The 7-[NAME_20] of Action (Norman 1988)

Predicting vs.

- Most goals not satisfied by a single action.
- Numerous sequences - discoverability
- May span seconds or minutes or hours or days.
- Continuous feedback
 - results may spawn other goals and other actions
 - goals lead to sub-goals
 - intentions lead to sub-intentions
- In a large activity, intermediate goals can be forgotten, discarded, or reformulated)



The 7-¹ of Action (Norman 1988)

Predicting vs. ²

Gulf of ³ ⁴ ⁵ ⁶ ⁷ ⁸ ⁹ ¹⁰ ¹¹ ¹² ¹³ ¹⁴ ¹⁵ ¹⁶ ¹⁷ ¹⁸ ¹⁹ ²⁰ ²¹ ²² ²³ ²⁴ ²⁵ ²⁶ ²⁷ ²⁸ ²⁹ ³⁰ ³¹ ³² ³³ ³⁴ ³⁵ ³⁶ ³⁷ ³⁸ ³⁹ ⁴⁰ ⁴¹ ⁴² ⁴³ ⁴⁴ ⁴⁵ ⁴⁶ ⁴⁷ ⁴⁸ ⁴⁹ ⁵⁰ ⁵¹ ⁵² ⁵³ ⁵⁴ ⁵⁵ ⁵⁶ ⁵⁷ ⁵⁸ ⁵⁹ ⁶⁰ ⁶¹ ⁶² ⁶³ ⁶⁴ ⁶⁵ ⁶⁶ ⁶⁷ ⁶⁸ ⁶⁹ ⁷⁰ ⁷¹ ⁷² ⁷³ ⁷⁴ ⁷⁵ ⁷⁶ ⁷⁷ ⁷⁸ ⁷⁹ ⁸⁰ ⁸¹ ⁸² ⁸³ ⁸⁴ ⁸⁵ ⁸⁶ ⁸⁷ ⁸⁸ ⁸⁹ ⁹⁰ ⁹¹ ⁹² ⁹³ ⁹⁴ ⁹⁵ ⁹⁶ ⁹⁷ ⁹⁸ ⁹⁹ ¹⁰⁰ ¹⁰¹ ¹⁰² ¹⁰³ ¹⁰⁴ ¹⁰⁵ ¹⁰⁶ ¹⁰⁷ ¹⁰⁸ ¹⁰⁹ ¹¹⁰ ¹¹¹ ¹¹² ¹¹³ ¹¹⁴ ¹¹⁵ ¹¹⁶ ¹¹⁷ ¹¹⁸ ¹¹⁹ ¹²⁰ ¹²¹ ¹²² ¹²³ ¹²⁴ ¹²⁵ ¹²⁶ ¹²⁷ ¹²⁸ ¹²⁹ ¹³⁰ ¹³¹ ¹³² ¹³³ ¹³⁴ ¹³⁵ ¹³⁶ ¹³⁷ ¹³⁸ ¹³⁹ ¹⁴⁰ ¹⁴¹ ¹⁴² ¹⁴³ ¹⁴⁴ ¹⁴⁵ ¹⁴⁶ ¹⁴⁷ ¹⁴⁸ ¹⁴⁹ ¹⁵⁰ ¹⁵¹ ¹⁵² ¹⁵³ ¹⁵⁴ ¹⁵⁵ ¹⁵⁶ ¹⁵⁷ ¹⁵⁸ ¹⁵⁹ ¹⁶⁰ ¹⁶¹ ¹⁶² ¹⁶³ ¹⁶⁴ ¹⁶⁵ ¹⁶⁶ ¹⁶⁷ ¹⁶⁸ ¹⁶⁹ ¹⁷⁰ ¹⁷¹ ¹⁷² ¹⁷³ ¹⁷⁴ ¹⁷⁵ ¹⁷⁶ ¹⁷⁷ ¹⁷⁸ ¹⁷⁹ ¹⁸⁰ ¹⁸¹ ¹⁸² ¹⁸³ ¹⁸⁴ ¹⁸⁵ ¹⁸⁶ ¹⁸⁷ ¹⁸⁸ ¹⁸⁹ ¹⁹⁰ ¹⁹¹ ¹⁹² ¹⁹³ ¹⁹⁴ ¹⁹⁵ ¹⁹⁶ ¹⁹⁷ ¹⁹⁸ ¹⁹⁹ ²⁰⁰ ²⁰¹ ²⁰² ²⁰³ ²⁰⁴ ²⁰⁵ ²⁰⁶ ²⁰⁷ ²⁰⁸ ²⁰⁹ ²¹⁰ ²¹¹ ²¹² ²¹³ ²¹⁴ ²¹⁵ ²¹⁶ ²¹⁷ ²¹⁸ ²¹⁹ ²²⁰ ²²¹ ²²² ²²³ ²²⁴ ²²⁵ ²²⁶ ²²⁷ ²²⁸ ²²⁹ ²³⁰ ²³¹ ²³² ²³³ ²³⁴ ²³⁵ ²³⁶ ²³⁷ ²³⁸ ²³⁹ ²⁴⁰ ²⁴¹ ²⁴² ²⁴³ ²⁴⁴ ²⁴⁵ ²⁴⁶ ²⁴⁷ ²⁴⁸ ²⁴⁹ ²⁵⁰ ²⁵¹ ²⁵² ²⁵³ ²⁵⁴ ²⁵⁵ ²⁵⁶ ²⁵⁷ ²⁵⁸ ²⁵⁹ ²⁶⁰ ²⁶¹ ²⁶² ²⁶³ ²⁶⁴ ²⁶⁵ ²⁶⁶ ²⁶⁷ ²⁶⁸ ²⁶⁹ ²⁷⁰ ²⁷¹ ²⁷² ²⁷³ ²⁷⁴ ²⁷⁵ ²⁷⁶ 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The 7-¹ of Action (Norman 1988)

Predicting vs. ²

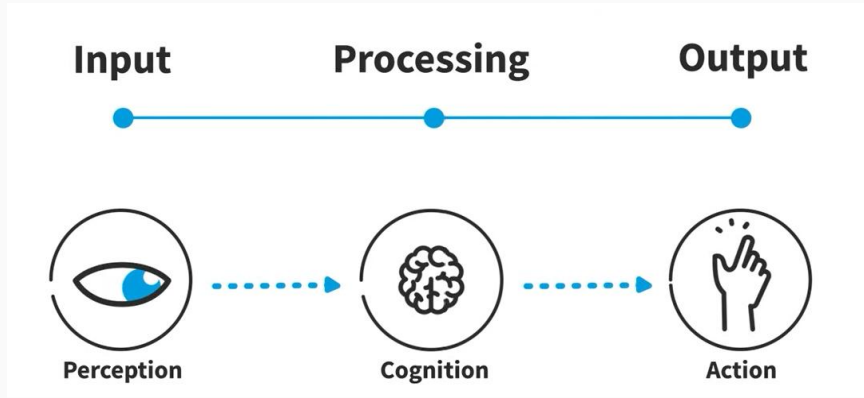
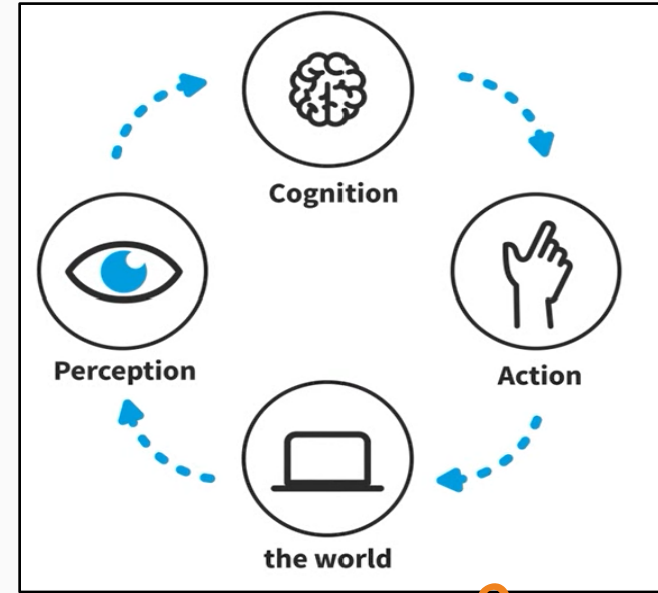
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Perceptions & [NAME_46]

[NAME_46] sensation

Key for constructing understanding of the world

A process through which individuals construct a mental representation of the external world, enabling them to think about it and effectively act within it. This concept extends beyond the mere reception of sensory inputs through our sense organs; it encompasses the complex interplay between cognitive processes and sensory systems to make sense of the world, especially when interacting with digital interfaces.



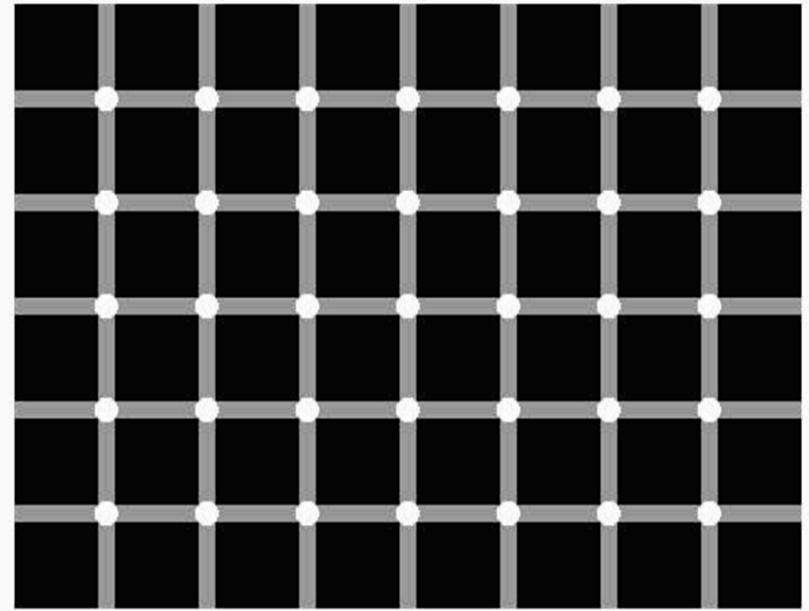
Perception vs Perspective

Perceptions & [NAME_46]

[NAME_46] sensation

Key for constructing understanding of the world

Illusions are special perceptual experiences in which information arising from “real” external stimuli leads to an incorrect perception, or false impression, of the object or event from which the stimulation comes



Knowledge about the physical properties of objects cannot be gained by perception, so perception is neither “veridical” nor “valid” in the strict sense of the words—the properties of the “thing in itself” remain indeterminate in any empirical sense ([Kant, 1787/1998](#))

- They tell us the limits and capacity of our perceptual apparatus
- They can specify how the constraints of perception are set



[NAME_47]

[NAME_47] can a person do with the things shown here?

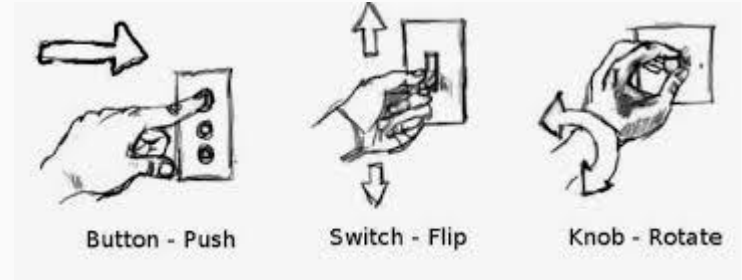


Affordance, reception & time

Why was the door designed this way?

Affordances: Expression often used unclearly

Especially in the field of usability, “affordance” is an expression often used only semi-correctly. Change in meaning over time:



Affordances: Expression often used unclearly

Especially in the field of usability, “affordance” is an expression often used only semi-correctly. Change in meaning over time:

